

Stock Market Returns and Portfolio Construction Using Graph Attention Networks

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Abstract—In the research of financial market analysis, the correlation research and portfolio optimization has important theoretical value, but financial industry players (hedge funds, HFTs, Investment Banks) still rely on traditional statistical and machine learning approaches to predict the market. In this report, a comprehensive review of how co-movement between stocks can be exploited to generate more returns than the benchmark returns by using a deep learning technique Graph Attention Networks has been discussed. A Graph Attention Network with fine tuned hyperparameters was built to predict the stock movement. The dataset involved data of 24 Indian stocks from year 2018 to 2024. Financial datasets tend to have more noise rather than signals, so 12 technical indicators were added as features to input to the Graph Attention Network. Each stock represents a node of a graph and edges of the graph represent the correlation between those two particular stocks through which the edge has been connected. So if the correlation between the two stocks has been greater than a threshold value, those two stocks are connected with an edge and the edge represent the correlation value between those two stocks. The Graph Attention Network was trained and carefully tuned hyperparameters help achieve a directional accuracy of 58.69%. The model predicts 21-day forward returns. Finally the portfolio construction strategy involved picking up 12 top performing stocks for the next 21 days predicted by the model and repeat the sequence every 21 days. The strategy generated an alpha of 5.6% on 2023 data and 4.1% on 2024 data.

Keywords: Alpha (the excess returns that a stock market strategy generates over ordinary market returns)

I. INTRODUCTION (HEADING I)

Stock market prediction is very challenging. Financial industry professionals still use traditional statistical and machine learning tools to predict a stock market, treating each stock independently. But instead stocks tend to move together, stocks tend to have a correlation between each other. For example, IT stocks move together, banking stocks move together. The problem arises when we try to capture these relationships. Linear regressive statistical models can't learn complex patterns of a financial markets. Stock markets have very different and complex patterns, for example some of them exhibit a trend, few of them exhibit seasonal patterns, many of them exhibit co-movement and are correlated to each other. Traditional statistical tools fail are not robust to these patterns. So here is a novel approach to capture the relationships, Graph Attention Networks. The network of graphs is built of nodes and edges, where each node represents a stock and each edge represents the correlations between the stocks. The objective of the approach is to build a Graph Attention Network for the selected stocks, create a portfolio strategy and beat the benchmark returns of an equal weighted portfolio. Deep learning techniques are often robust, captures

essential patterns and often beat the statistical methods if tuned properly. A graph attention network captures engineered features of each stock data and correlations between each of the selected 24 stocks.

II. LITERATURE REVIEW

A. Traditional approaches

Statistical tools like ARIMA, LSTM are used for time series analysis and forecasting which are not robust enough to capture the hidden signals among stocks and inside the stock. These models treat each stock separately and end up capturing only trend and seasonal components of the time series. But the correlation among the stocks carry heavy sentiments about the market direction. These models fail to capture the correlation info among multiple stocks.

B. Recent Work

There has been many approaches to predict market returns by using other deep learning techniques like using machine learning models, Graph Neural Networks. There has been work on stock co movement prediction using the technical indicators. But the Graph Attention Network has been most unexplored and untouched ones. This is something which makes this deep learning technique a novel approach to portfolio optimization strategies. There has been a subtle gap and limited work in this area in the Indian market. This is something which makes this area a research worthy. Currently there has been no Graph Attention Network based portfolio strategies for National Stock Exchange stocks.

III. METHADAOLOGY

This section involves a detailed methadology starting from the data of 24 stocks, GAT model training, hyperparameter tuning, portfolio strategy and final results.

Section 3.1: Data

The dataset involves the data of handpicked 24 stocks listed under National Stock Exchange India. The total dataset is the data of all stocks from year 2018-2024. The whole dataset has been sourced from Yahoo finance API. The initial features involved Open, High, Low, Close and Volume of the stocks traded each day for these 7 years. The data has been splitted carefully to train set (5 years of data from year 2018-2022), validation set (1 year of data of year 2023), and test set (1 year of data of year 2024). The data has been carefully studied and splitted in order to avoid any type of data leakage among these three sets generated out of the whole dataset

Section 3.2: Feature Engineering

This is the most crucial section of the project. The current data is more of a noise rather than predictive signals which a Graph

Attention Network could capture. To ensure this, feature engineering has been done on the financial datasets which involve adding technical indicators to the data. So for each stock, the following technical indicators were picked and added to the features of each stock. as listed below:

- EMAs (Exponential Moving Averages): A weighted moving average which gives more weight to recent prices to capture short term trends in the market.
- VWAP (Volume Weighted Average Price): Represents average traded price of a stock during the day weighted by trade volume.
- MACD (Moving Average Convergence Divergence): Measures momentum of a stock price by comparing short term and long term EMAs.
- ADX (Average Directional Index): Quantifies the strength of a stock trend.
- ATR (Average True Range): Average of a true range of a stock.
- Bollinger Band Width: Indicates volatility of a stock by measuring the distance between upper band (mean + 2 standard deviations) and lower band (mean - 2 standard deviations).
- Statistical Volatility: Standard deviation of returns.
- RSI (Relative Strength Index): an oscillating technical indicator which takes volume and momentum of stock into the account.
- ROC (Rate of Change): Shows momentum by using the percentage change in prices compared to a previous period.
- EMA slope: Represents the slope of Exponential Moving Averages.
- Donchian Width: Measures the difference between highest high and lowest low of a stock prices

These features were calculated using the technical analysis module provided by Python and the original OHLCV data was scraped and replaced by these calculated features. Further to keep each feature in the same range, a Standard Scaler was fitted on the training set and dataset has been transformed with a normalized feature data.

Section 3.3: Graph Construction

The nodes of the graph represents the selected 24 stocks while the node features are the technical indicators feature engineered in the previous section. The most important and crucial part of a graph are edges which were created based on 60-day rolling correlation among the 24 selected stocks. Finally, a threshold correlation to get an edge among two stocks was decided so that our model actually capture predictive signals rather than noisy correlations. All the stocks with absolute value of correlation among them more than 0.15 get an edge between them. The edge weight here is the strength of correlation between the stocks. And since these correlations were decided on a rolling window of 60 days, the correlation among nodes updates daily and thus this is a dynamic graph.

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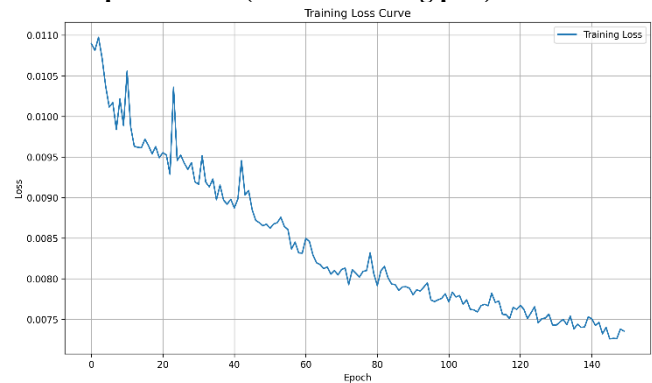
Section 3.4: GAT Model Architecture

Multiple Graph Attention Networks were trained on a wide hyperparameter grid, each being trained robustly on the 5 year data of 2018-2022. Based on the results and performance of each model, the model with these architecture parameter was finalized:

- Input : 12 features per node
- No. of GAT layers: 2
- Hidden Dimension: 32
- Attention heads : 8
- Dropout : 0.25
- Output : 1 prediction per stock
- Loss : Mean Squared Error
- Optimizer : AdamW
- Learning rate : 5e-4
- Epochs : 150
- Framework : Pytorch geometric

The output embeddings of each node inputsto a Multi layer perceptron which predict the 21 day ahead market returns of the stock one at a time. This model achieved a directional accuracy of 58.69% and was successful in capturing the variance of the actual returns.

Plot I: Epoch vs loss (Model Training plot)



Section 3.5: Portfolio Strategy

Finally after all the robust training and evaluation of multiple models comes the portfolio construction part. The GAT model outputs the 21 day ahead market returns of each stock for every training point. So two different portfolios were formed each being allocated a capital of INR 1,00,000. Portfolio A was formed with the top performing 12 stocks out of 24 stocks predicted by the GAT model. Portfolio B was formed which equally weighs all 24 stocks. The strategy was that the portfolio A rebalances every 21 days and add a new set of 12 top performing stocks predicted by the Graph Attention model (21 day ahead returns) on that day. So the portfolio A rebalances 12 times every year as each month has 21 trading days

IV. RESULTS

Section 4.1: Model Performance

The model performed with a directional accuracy of 58.69% on the 2023 validation set. The directional accuracy of 58.69% suggests that the model actually captured predictive signals rather than noise from the data. Generally, a directional accuracy of 50% is considered as random. The model was successful in capturing the sentiments of the

financial markets. The model gave a mean squared error of 0.0056 on 2023 validation set and a standard deviation ratio between model predictions and actual returns turned out to be 0.70 which suggests that the model was neither too conservative to underfit nor too flexible to overfit on the training set. The model reasonably captured the variance of returns.

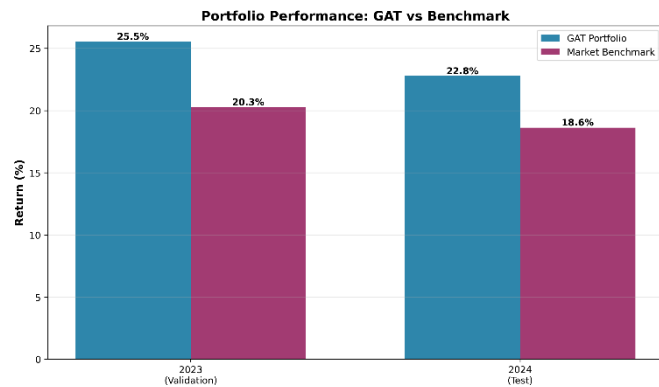
Section 4.2: Backtesting results

Table I: Portfolio performance

Year	GAT-based Portfolio Return (Portfolio A)	Benchmark Return (Portfolio B)	Alpha
2023	25.55%	20.27%	5.28%
2024	22.78%	18.62%	4.16%

The top 12 stock selecting strategy outperformed the benchmark returns by 5.28% on 2023 year validation data and by 4.16% on 2024 year test data. This means the strategy gave excess returns across both years of back testing the strategy. The average alpha generated was 4.85%.

Plot II: Portfolio A vs Portfolio B



Section 4.3: Further Analysis

Further, to check the robustness of the results, Portfolio A which constituted of top 12 performing stocks was then backtested with a different portfolio sizes except 12. The portfolio A was constructed with top 10 performing stocks rather than top 12 and when backtested generated an alpha of 3.82% on year 2023 validation data and 2.97% on year 2024 test data. For more analysis, top 15 stocks were selected and again it gave excess returns on the market. The top 15 stock strategy generated an alpha of 3.27% on year 2023 validation data and 1.96% on year 2024 test data.

Table II: Portfolio Performance with top 10 stock strategy

Year	GAT-based portfolio return (Portfolio A)	Benchmark Return (Portfolio B)	Alpha
2023	24.1%	20.27%	3.82%

2024	23.24%	18.62%	2.97%
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Table III: Portfolio Performance with top 15 stock strategy

Year	GAT-based portfolio return (Portfolio A)	Benchmark Return (Portfolio B)	Alpha
2023	23.54%	20.27%	3.27%
2024	20.58%	18.62%	1.96%

If observed carefully, the alpha generated on 2023 validation set is slightly lower than the 2024 test set, which clearly implies that the model has learnt the predictive signals. In finance industry, the alpha generally decays as a strategy is being tested on multiple years.

V. CONCLUSION

The project successfully demonstrated how Graph Attention Networks can exploit correlation structure among a network of stocks to generate excess returns and beat the market.

Section 5.1: Key Findings

- GAT model achieved 58.69% directional accuracy in predicting 21-day forward returns.
- Portfolio strategy selecting top 12 predicted performers generated consistent outperformance
- 2023 validation: 5.28%
- 2024 test: 4.16%
- Average alpha: 4.72%

Section 5.2: Significance

The positive alpha across both validation and test sets shows that model has learned genuine patterns in stock co-movements rather than overfitting to the training set

Section 5.3: Future Work

- The work can be extended towards larger pool of stocks.
- Additional data sources, risk free rates and transaction costs can be incorporated.

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